

Illya Yurchanka

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M.Sc. Physics student at the University of Warsaw with strong expertise in LLM accuracy evaluation, benchmarking, and inference optimisation. Implemented Multi-Head Latent Attention (MLA) for memory-efficient LLM inference and developed end-to-end ML pipelines with PyTorch. Extensive experience in ROC/AUC analysis, benchmark development, and performance evaluation of large language models. Seeking to contribute to state-of-the-art LLM evaluation infrastructure at NVIDIA.

SKILLS

ML & AI: LLMs (MLA, KV Cache), Transformers, NLP, Vector Embeddings, CNN, RNN, Supervised/Unsupervised Learning, Hyperparameter Optimisation, LLM Accuracy Evaluation, ROC/AUC Analysis, Benchmark Development, Performance Benchmarking

Programming: Python (PyTorch, TensorFlow, Scikit-Learn, Pandas, NumPy, Matplotlib, SciPy), C++ (STL, Eigen, Boost), SQL (DuckDB, PostgreSQL)

Tools & DevOps: Git, Docker, Linux/Bash, CMake, Make, ROOT/TMVA, PyTest, Pydantic, Mathematica

Languages: English C1, Polish C1, Russian (Native), Belarusian (Native)

EDUCATION

University of Warsaw | *M.Sc. in Physics* **Oct 2025 – Jun 2027 (expected)**

- GPA: 4.5/5 Focus: Particle Physics, Machine Learning applications, Statistical Data Analysis

University of Warsaw | *B.Sc. in Physics* **Oct 2021 – Jun 2025**

- GPA: 4.5/5 Thesis: Identification of Λ hyperons in CBM spectrometer using TMVA (ML)

EXPERIENCE

M.Sc. Researcher – CBM Experiment, GSI/FAIR (Darmstadt) **Oct 2025 – Present**

- Developing C++ data processing workflows for the future CBM detector, applying ML techniques to simulated Au+Au collision data to improve particle identification and background rejection.
- Performing quantitative analysis of model performance against validation benchmarks; evaluating classification accuracy and optimising detection thresholds using ROC/AUC metrics.
- Managing and validating large simulation datasets in a production-like scientific computing environment; maintaining pipeline integrity across distributed analysis workflows.
- Collaborating with an international team across Warsaw and GSI Darmstadt, coordinating codebase updates and contributing to shared analysis infrastructure.

Research Intern – General Relativity **Mar 2025 – Jun 2025**

Center for Theoretical Physics, Polish Academy of Sciences

- Conducted independent research on general relativity; reviewed 10+ peer-reviewed publications to identify novel theoretical approaches and computational methodologies.
- Built mathematical models and performed symbolic/numerical computations in Mathematica and SciPy, producing analytical results presented to senior researchers.

Warsaw Econometric Challenge **Apr 2025**

- Led a team in a data science competition; rapidly adopted SQL and DuckDB frameworks to deliver ML-based predictive solutions under strict time pressure.
- Applied Scikit-Learn pipelines for competitive modelling, gaining practical end-to-end deployment experience; delivered validated predictions against held-out test sets.

PROJECTS

Multi-Head Latent Attention Implementation

Oct 2025

Python, PyTorch [GitHub]

- Implemented the MLA mechanism from scratch, compressing the KV cache by projecting keys and values into a low-dimensional latent space – significantly reducing inference memory footprint while preserving attention quality on par with standard multi-head attention.
- Conducted thorough benchmarking and comparative evaluation of attention mechanisms, quantifying accuracy and efficiency trade-offs using standard LLM benchmarks.

Food Nutrition Analysis with Machine Learning

Jan 2026

Python, Scikit-Learn, Pandas, NumPy, Transformers [GitHub]

- Built an end-to-end ML pipeline on the Open Food Facts dataset: predicted NutriScore grades, estimated calorie content, classified ultra-processed foods (NOVA), and scored environmental impact (Eco-Score).
- Implemented a semantic search recommender using vector embeddings to surface healthier product alternatives.

Bachelor Thesis – Λ Hyperon Identification in CBM Spectrometer

May 2025

C++, ROOT, ROOT-TMVA

- Benchmarked KNN, AdaBoost, and MLP classifiers for signal/background separation in high-energy physics data; ensemble methods achieved the best ROC-AUC (0.95+), directly informing the experiment's offline reconstruction strategy.
- Managed the full evaluation pipeline from raw data ingestion to validated results across a large simulation dataset; developed automated benchmark scripts for model comparison.

RELEVANT COURSES

Machine Learning Neural networks (FC, CNN, GAN, VAE), Transformers, RNNs, NLP, NumPy/Pandas/Matplotlib
2023/2024

Statistical Analysis of Experimental Data Statistical inference, hypothesis testing, uncertainty estimation, ROC/AUC evaluation, model benchmarking, data interpretation 2025/2026

Programming & Numerical Methods Advanced C++, numerical methods, physical simulations in Python 2023/2024

Programming STL algorithms, templates, containers 2022/2023